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PI Name:	Long, Marshall Ph.D.		
Project Title:	Coflow Laminar Diffusion Flame		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
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No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: change in Period of Performance to 1/1/2012-12/31/2014 per NSSC information (previous POP was 10/1/11-9/30/14)--Ed., 12/30/2011		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Smooke, Mitchell Ph.D. (Yale University)		
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Task Description:	Editor's Note: Successor agreement to Fuel Dilution Studies of Coflow Laminar Diffusion Flames in a Microgravity Environment (NNX07AB33A), per D. Stocker, NASA Glenn Research Center. Diffusion flames are the flame type of most practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in such systems is critical in the modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation and radiative transfer take place. Turbulent diffusion flames are generally nonbuoyant due to the large flow velocities. In addition, in practical devices combustor configuration and flow field patterns may produce flames with sharp edges. Extinction along these edges can result from incomplete combustion. In other regions combustion generated soot particulates may be formed. To understand better the factors that affect flame extinction and particulate formation, we propose to focus this study on flame phenomena at the extremes of the fuel dilution spectrum – namely dilute flames near extinction and pure-fuel (sooting) flames. Previous work has shown that existing computational models are unable to predict accurately the behavior of these systems. The enhanced time and spatial scales in a microgravity environment will enable the stabilization of flames with increased levels of dilution and will help generate flames that have increased soot residence times compared to normal gravity facilities. This should provide an excellent platform to gain a better understanding of the factors that affect diffusion flame lift-off distance and extinction, and it should also provide an excellent test-bed to determine the roles of inception, surface growth and oxidation processes in soot formation models. The overall goals of the research are to improve our understanding of the physical and chemical processes controlling diffusion (i.e., non-premixed) flame structure and lifting phenomena (i.e., stabilization) and to provide for rigorous testing of numerical models, including thermal radiation, soot formation, and detailed chemical kinetics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Combustion of fossil fuels is responsible for the majority of energy production in the world today and provides the power for transportation systems of all kinds. Effective modeling of combustion processes is critical to improving efficiencies and reducing pollutants in practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in diffusion flames will improve modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation and radiative transfer take place.
Task Progress:	Quantitative Measurements with Simple Diagnostics During the past year, we have continued the refinement and characterization of diagnostics and flame configurations that will allow us to take greatest advantage of the long periods of high-quality microgravity that are available on the International Space Station (ISS). We have worked to develop techniques for making the comparison between experiments and computations as quantitative as possible, even with relatively simple diagnostic techniques. A technique utilizing a color digital camera has been developed that allows observation of flame shape, size, lift-off height, and measurements of temperature via soot and thin-filament pyrometry as well as quantitative determination of the soot volume fraction. Techniques for obtaining an absolute light calibration have also been developed, which is essential for measuring the absolute soot volume fraction and the number density of chemiluminescent species such as CH*. Thin-filament pyrometry will be used in the Advanced Combustion in Microgravity Experiments (ACME) for temperature measurement. The technique has also been recently investigated and assessed against laser-based techniques. Several sources of error were identified and uncertainty analysis has been performed to estimate the measurement uncertainty and provide insights on error reduction. Microgravity Experiments: SLICE and CLD Flame Our Coflow Laminar Diffusion Flame experiment (CLD Flame) is one of the four investigations that make up ACME. The experiments in ACME are currently in development for conduct within the Combustion Integrated Rack (CIR) on the ISS. The CLD Flame hardware is currently being developed, and during the past year we have worked to improve the design of the coflow burner. A series of refinements were made to the fuel nozzle and the flow straightener used in the coflow. Detailed velocity measurements were made, which will provide the boundary conditions needed for the computational modeling. The improved coflow provided lifted flames with a higher degree of symmetry than previous designs. Following construction and characterization of the new burner at Yale, it was shipped to NASA Glenn for testing in their drop tower facilities. As a precursor to CLD Flame, the Structure & Liftoff in Combustion Experiment (SLICE) takes advantage of the existence of a closely related set of experiments recently completed on board the ISS. The goal of SLICE is to investigate the structure of lifted and sooty flames. The results can provide a test case for developing the computational model and soot sub-model, and also be used to minimize risk and maximize the scientific return of the CLD Flame Experiment. SLICE Experiments were carried out on board the ISS early in 2012. A comprehensive data set with a variety of test conditions, including different hydrocarbon fuels with varying dilution levels, different fuel nozzle sizes and various flow conditions, was obtained. A fully characterized color digital camera was used to image line-of-sight flame emissions to obtain flame properties such as the flame shape and size, lift-off height, as well as soot temperature and volume fraction. The data analysis that has been done includes quantitative determination of soot temperature and volume fraction for all sooty flames and lift-off height measurements for lifted flames. Similar tests using a copy of the SLICE hardware under normal gravity have also been performed that enable direct comparisons of these flame properties between 0-g flames and 1-g flames. Preliminary numerical simulation of the flames has also been performed and compared with experimental results. Computational modeling Recent computational efforts have concentrated on computing flames that were measured in SLICE to study the influences of gravity and fuel dilution on the structure, stabilization, and sooting tendency of laminar coflow methane-air diffusion flames. A series of nitrogen-diluted flames measured was assessed numerically under normal gravity and microgravity conditions with the fuel stream CH4 mole fraction ranging from 0.4 to 1.0. The vorticity-velocity formulation of the governing equations was employed to describe the reactive gaseous mixture, where the soot evolution process was considered as a classical aerosol dynamics problem and was represented by the sectional aerosol equations. Since each flame is axisymmetric, a two-dimensional computational domain was employed, where the grid on the axisymmetric domain was a nonuniform tensor product mesh. The governing equations and boundary conditions were discretized on the mesh by a nine-point finite difference stencil, with the convective terms approximated by a monotonic upwind scheme and all other derivatives approximated by centered differences. The resulting set of fully coupled, strongly nonlinear equations was then solved simultaneously at all points using a damped, modified Newton’s method and a nested bi-conjugate gradient stabilized (Bi-CGSTAB) linear algebra solver with a

	block-line Gauss-Seidel preconditioner. Due to the lack of buoyancy force, the 0-g flames have much less axial velocity and more residence time as verified by the numerical simulations. The microgravity flames were found to be sootier than normal gravity flames by almost one order of magnitude in terms of the peak soot volume fraction. The distribution of soot was also markedly different. For several conditions studied, the peak soot concentration occurred on the flame centerline under normal gravity and on the wings under microgravity conditions. The flame temperature in 0 g is lower than its 1-g counterpart due to higher soot loading and more radiation loss. Quantitative comparisons between experimental results and numerical simulations show reasonable agreement on flame shape, temperature, and soot volume fraction in normal gravity. The computational model also successfully captured the difference in location of the peak soot volume fraction – on the centerline at 1 g and on the wings at 0 g. However, accurate prediction of soot volume fraction in 0 g remains a challenge as the computed peak value is smaller than the measured peak value by an order of magnitude. More advanced reabsorption submodels will be introduced, and the soot surface growth coefficient will be adjusted to improve the accuracy of the predictions.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
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